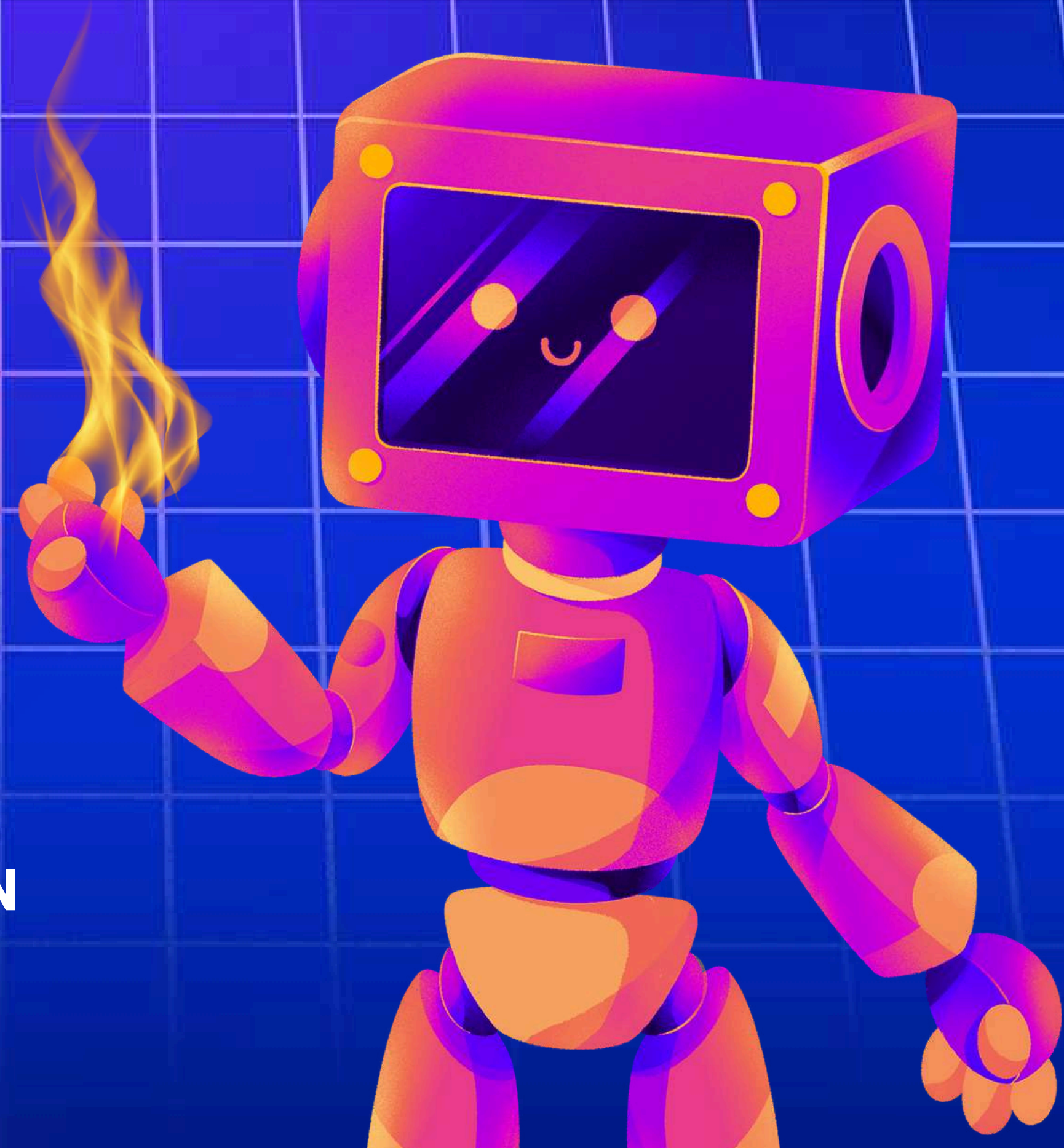


TEAM 44

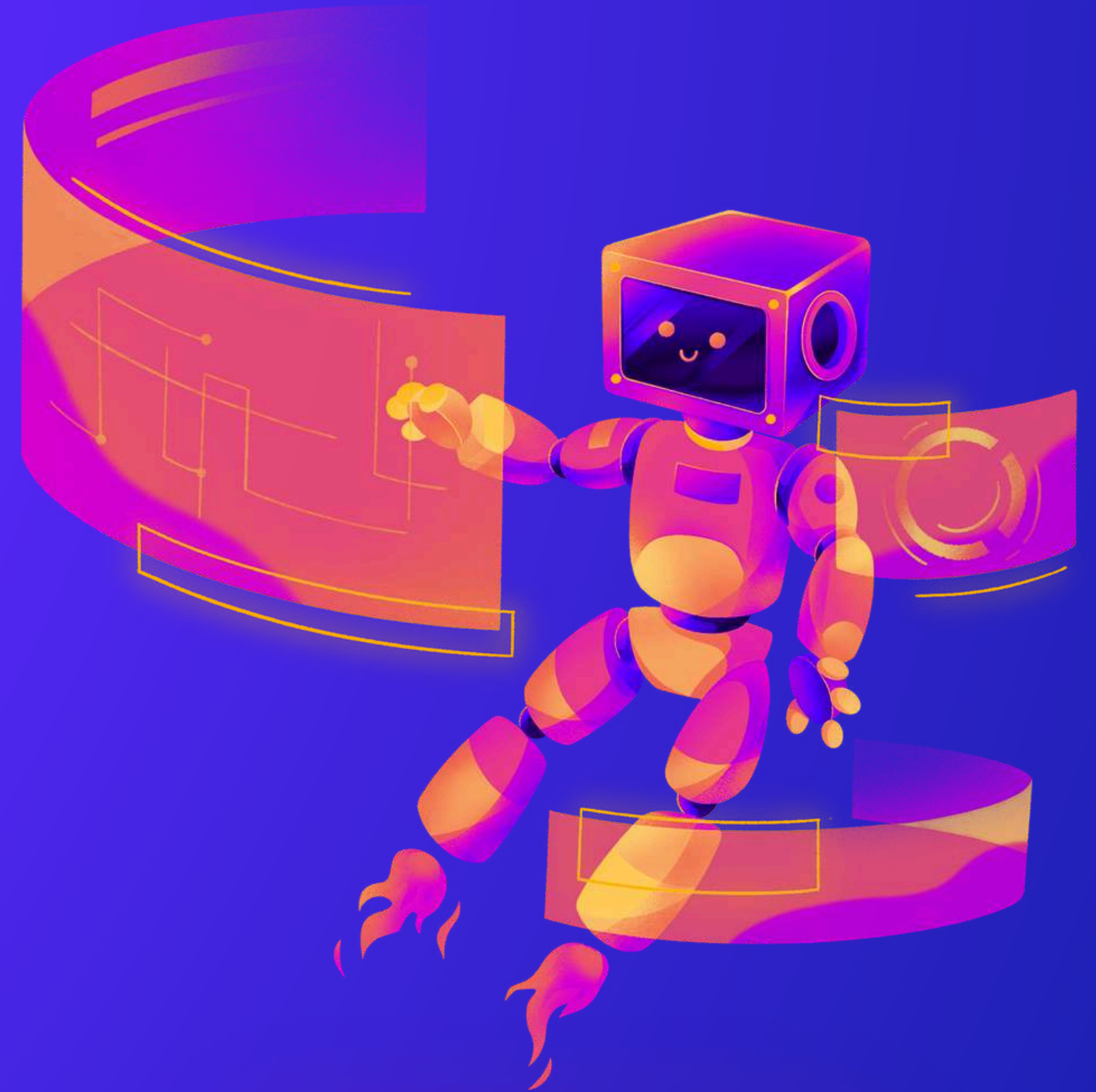
FIRE EXTINGUISHING ROBOT

**FIRE EXTINGUISHING AND
NAVIGATING SYSTEM USING AN
ARDUINO UNO**



TEAM MEMBERS:

	USN	NAME
1	070	Anusree Krishnan
2	003	Aadya Arun Talreja
3	073	Archita Das
4	075	Arjun Syam
5	269	Misha Phokela
6	273	Mitul KM



ROBOTICS

DOMAIN LEAD - HONGRAY SIR

PROBLEM STATEMENT

DETECT AND RESPOND TO FIRE EMERGENCIES BY DESIGNING AND DEVELOPING A FIRE DETECTION AND NAVIGATION SYSTEM. IN ORDER FOR THE SYSTEM TO BE EFFECTIVE, IT SHOULD BE CAPABLE OF IDENTIFYING THE PRESENCE OF FIRE, LOCATING ITS SOURCE, AND PUTTING IT OUT AS QUICKLY AS POSSIBLE.

INTRODUCTION TO OUR ROBOT

FIREFIGHTER ROBOT CAN BE USED TO CONTROL THE FIRE. THIS ROBOT CAN DETECT THE FIRE BY ITSELF AND CONTROL THE FIRE BY THROWING WATER. THERE ARE SOME SENSORS WE ARE USING THAT CAN DETECT FIRE AND ROBOTS CAN MOVE THERE TO FIRE EXTINGUISH.

FIRE DETECTION TECHNOLOGIES

SMOKE DETECTORS, WHICH USE EITHER IONISATION OR PHOTOELECTRIC PRINCIPLES, CAN DETECT SMOKE PARTICLES. SMALL PARTICLES PRODUCED BY FLAMING FIRES ARE MORE EASILY DETECTED BY IONISATION DETECTORS, WHILE LARGER PARTICLES PRODUCED BY SMOULDERING FIRES ARE MORE EASILY DETECTED BY PHOTOELECTRIC DETECTORS, THEY ARE PARTICULARLY USEFUL IN ENVIRONMENTS WHERE SMOKE DETECTION MAY NOT BE RELIABLE.

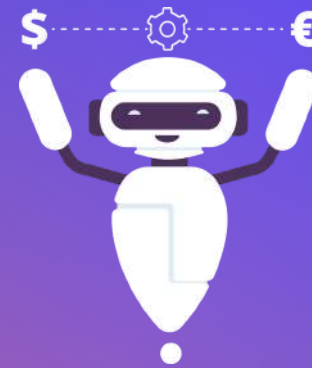
FIREFIGHTING TECHNIQUES AND EXTINGUISHING AGENTS

THRESHOLD-BASED ALGORITHMS TRIGGER ALARMS WHEN THE THRESHOLDS ARE EXCEEDED, WHILE MACHINE LEARNING ALGORITHMS IDENTIFY FIRE EVENT PATTERNS AND ADAPT TO DIVERSE CONDITIONS. FUSION ALGORITHMS INTEGRATE DATA FROM MULTIPLE SENSORS TO ENHANCE DETECTION ACCURACY AND RELIABILITY BY COMBINING INFORMATION FROM VARIOUS SENSOR TYPES.

WORKING



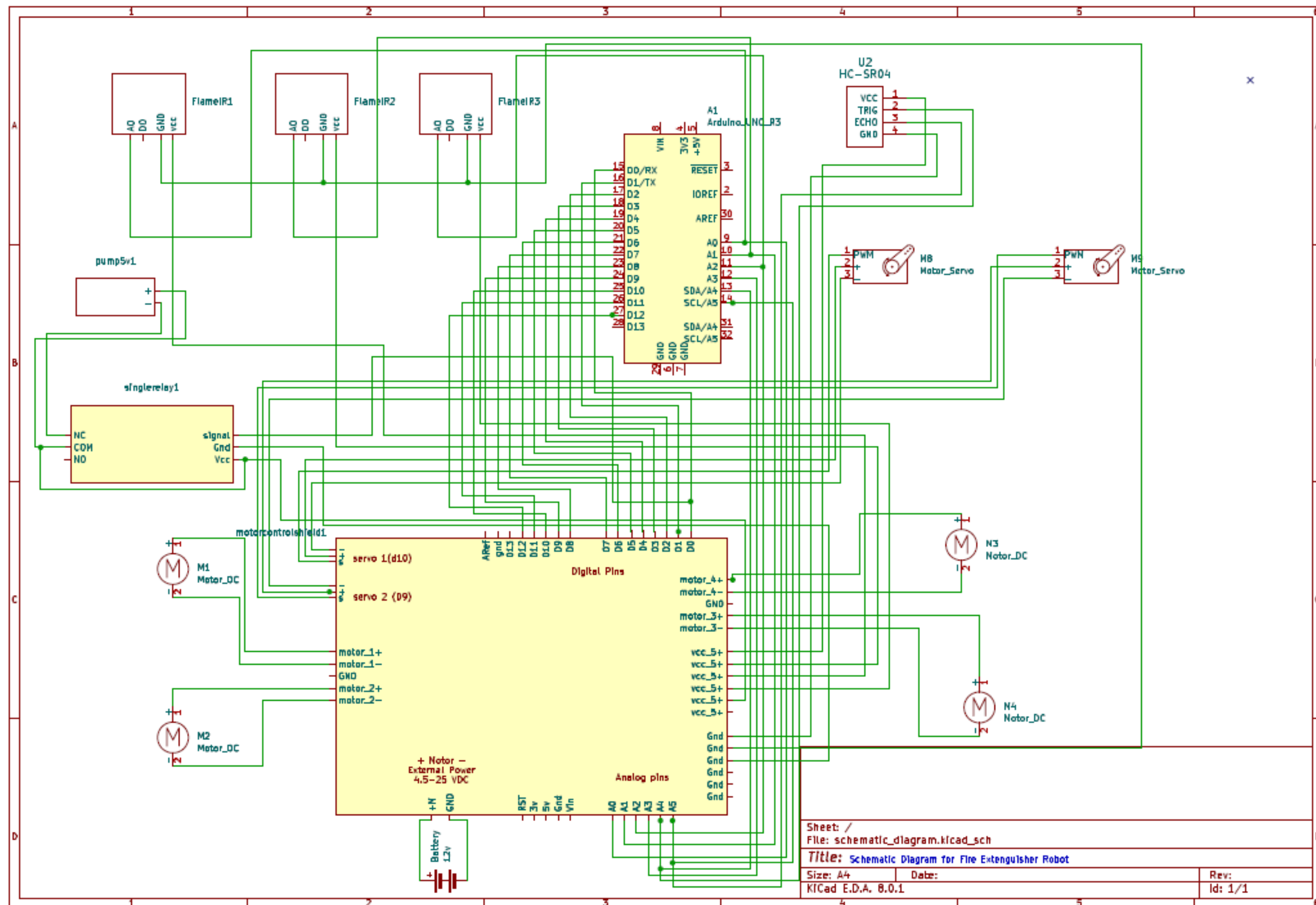
The basic working of the Firefighter robot is to detect an area of fire within a closed environment, smartly navigating to the area and extinguishing the fire. To achieve this the robot uses Arduino providing the algorithm to implement it.



Once the heat area is detected the Arduino will then give commands to the robot to navigate towards it avoiding any obstacles in the way



Once the bot is close enough to the area the Arduino will then send the command to spray water/CO2 to mitigate the damage.



OBJECTIVES

1

**DEVELOPING
ROBUST FIRE
DETECTION
CAPABILITIES**

2

**CREATING AN
EFFICIENT
NAVIGATION
SYSTEM**

3

**ENABLING
AUTONOMOUS
FIRE
SUPPRESSION
MECHANISMS**

4

**ENSURING
SAFETY,
RELIABILITY,
AND
COMPLIANCE**

COMPONENTS

Estimated Cost = 2000/-



- 1 Arduino Uno
- 2 Flame IR Sensor
- 3 Nozzle for water
- 4 Water Pump 5V
- 5 Robot Chassis
- 6 Single Channel Relay
- 7 12V Battery
- 8 Wires

TIMELINE



CONCEPTUALIZATION
AND DESIGN
(WEEK 1 - WEEK 2)



PROTOTYPING
(WEEK 2 - WEEK 4)



PROGRAMMING AND
SOFTWARE DEVELOPMENT
(WEEK 4 - WEEK 6)



TESTING
AND OPTIMIZATION
(WEEK 6 - WEEK 8)



FINE-TUNING AND
DEBUGGING
(WEEK 8 - WEEK 9)



DOCUMENTATION
(WEEK 9)



TRAINING
(WEEK 9)



DEPLOYMENT
(WEEK 10)

EXPECTED OUTCOMES:
TO HAVE A FUNCTIONAL FIRE EXTINGUISHING ROBOT

IMPACT:
ENHANCED SAFETY
24/7 AVAILABILITY
SCALABILITY AND ADAPTABILITY

APPLICATIONS

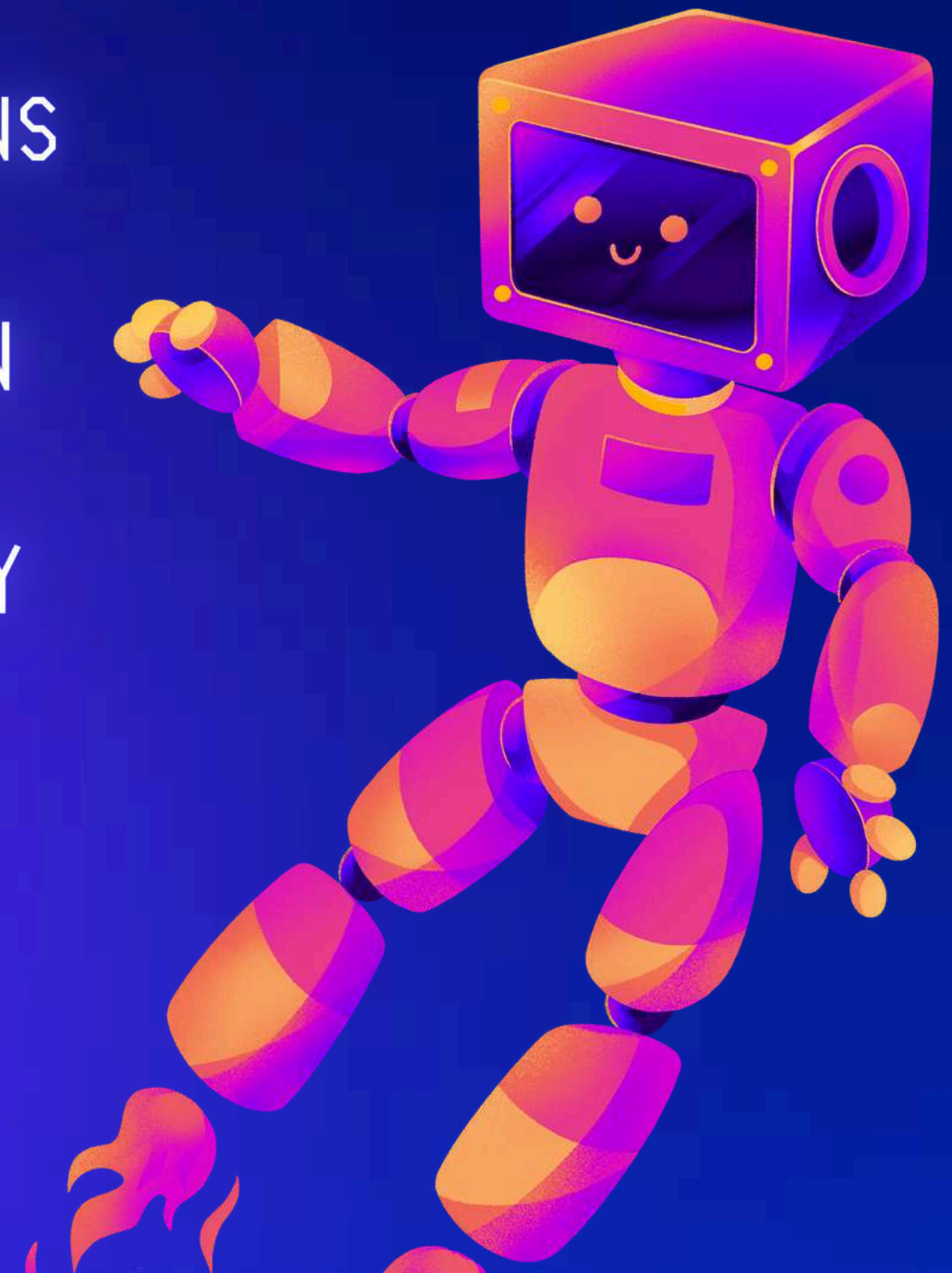
INACCESSIBLE ENVIRONMENTS

INDUSTRIAL KITCHENS

MAZE NAVIGATION

RESTAURANT SAFETY

MILITARY DEPLOYMENT



THANK YOU!

